

SPACE PRODUCTS **CATALOG**



RELIABLE MINIATURIZATION TECHNOLOGIES FOR ELECTRONICS

TABLE OF CONTENTS



ABOUT 3D PLUS	2
QUALITY ASSURANCE	4
SPACE EXPERTISE	5
CATALOG PRODUCTS	6
CUSTOM SOLUTIONS: SYSTEM-IN-PACKAGE	7
VOLATILE MEMORIES	8
NON VOLATILE MEMORIES	10
DDR4 ECOSYSTEM	12
TMR NOR FLASH MEMORIES	14
MASS DATA STORAGE MEMORIES	15
FUSIO RT COMPUTER CORES	16
CMOS CAMERA HEADS	18
INTERFACES	20
POINT-OF-LOAD CONVERTERS	21
PROTECTION ICs	22
DDR TERMINATION REGULATORS	23
FLIGHT HERITAGE	24

ABOUT 3D PLUS

COMPANY OVERVIEW

3D PLUS is a world leading supplier of advanced high density 3D microelectronic products and die/wafer stacking technology. Our products meet the demand for high reliability, high performance and very small size of today's and tomorrow's electronics.

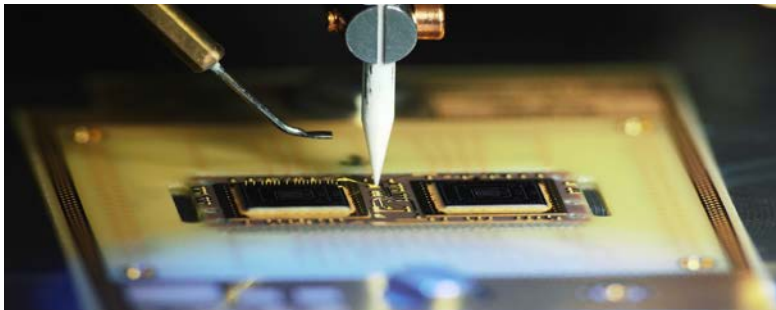
Our patented technology portfolio starts with standard package scale upward to die-size and wafer-level stacking processes. It enables stacking heterogeneous active, passive and opto-electronic devices in a single highly miniaturized package. Comparing to other existing 2D traditional solutions, our technology allows gaining a factor of at least 10 on the weight and volume of the components.

3D PLUS standard products and System-In-Package (SiP) solutions bring breakthrough advantages to our customers' electronic designs. They are used in diverse computer boards, data recorders boards and custom applications for buses and payloads. Our Flight Heritage is expanding continuously with products launched in Space every month in GEO, MEO and LEO orbits, for deep space exploration missions, for satellite constellation fleets, and for governmental missions.

Since our creation in 1995, we have been pursuing a constant growth with today customers spread over 30 countries.

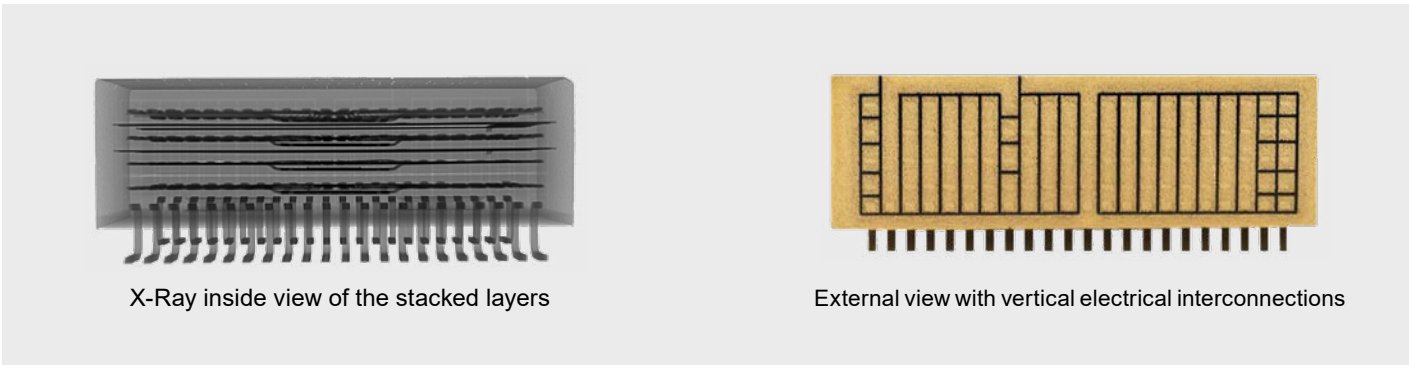


OUR UNIQUE KNOW-HOW



3D PLUS offers high reliability and high performance innovative products thanks to a strong design activity of complex electronic modules and a high level of technical capability in various areas such as digital, analog and IP CORES. These abilities, combined with a strong Space Radiation expertise, enable us to bring key advantages for all kinds of space missions with a very high level of guarantee.

3D PLUS has been recognized for its innovation in design and manufacturing of miniaturized 3D modules for more than 20 years. With a capability of stacking up to 10 semiconductor devices up to 1 mm each, 3D PLUS ultra low profile modules are unique.



3D PLUS MAIN BENEFITS



MINIATURIZATION

Size reduction for the PCB and weight savings for the application.



HIGH RELIABILITY

Proven technology combined with a unique design and manufacturing know-how developed for more than 20 years.



FLIGHT HERITAGE

"Design for Space" products definition. Space applications heritage upward to 20 years missions in geostationary orbit.



RADIATION EXPERTISE

A strong space radiation expertise for high reliability products. 3D PLUS offers to customers fully transparent solutions with high level of radiation performances for all Space missions.



PROVEN TECHNOLOGY - MANUFACTURING PROCESS

Our proven patented technology brings high performance and small footprint for customer applications. The European Space Agency approved 3D PLUS manufacturing line.



LONG TERM AVAILABILITY

Expertise in obsolescence management and long term supply, we can guarantee availability of our products for very long term programs.



HIGH PERFORMANCE

Extended thermal cycles - High mechanical resistance - High temperature - Space qualification.

QUALITY COMMITMENT

3D PLUS' Quality and Reliability Organization is the executive arm of the quality assurance for the entire company. It is responsible for ensuring that 3D PLUS has the most competitive systems, processes, and programs in place to drive strategic quality improvements goals. To ensure that stringent product quality and reliability expectations are integrated throughout the company, 3D PLUS has established a global Quality Policy. 3D PLUS will provide a competitive advantage to its customers through timely, innovative, and defect-free products with outstanding service and specific customer support organization with systematic continuous improvement.

ISO CERTIFICATION

Certification to external standards is one example of how 3D PLUS strives to meet the needs of its worldwide customers. 3D PLUS Headquarter factory, Technical Center and Sales sites, as well as our major subcontractors, are certified to prestigious internationally recognized quality standards. 3D PLUS has continued its Quality excellence path by successful periodic renewal audits performed by third-party Lloyd's Register Quality Assurance.

ESA CERTIFICATION

The European Space Agency approved 3D PLUS manufacturing line for Space Applications.

ENVIRONMENTAL POLICY

3D PLUS as ecoresponsible company is committed for several years to the environmental protection, of the goods and people. In its leading position of micro-electronic components manufacturer for space applications, the environment and the safety constitute major axes of improvement. 3D PLUS is engaged in the GREEN initiatives and committed to be in line with regulation providing statement concerning RoHS, REACH and Halogen Content.

SPACE EXPERTISE

SPACE APPLICATIONS

3D PLUS products and SiPs are used by the major Aerospace Prime contractors, systems and equipments manufacturers and space agencies worldwide. Recognized for their electrical performance, miniaturization, quality, reliability and radiation assurance level, 3D PLUS space qualified products are used in all types of space applications:

- **Consumer Applications:** telecommunication, navigation, Internet
- **Sustainable Development:** environment and climate monitoring
- **Defense & Security:** Earth observation
- **Space transportation:** launch and manned space vehicles
- **Science:** astronomy, deep space exploration and interplanetary missions



RADIATION ASSURANCE

With more than 175,000 modules in space as of 2021, and with more than 20 years of flight heritage with no reported failure, 3D PLUS is the largest Hybrid space qualified catalog products and customer system-in-packages (SIPs) manufacturer in Europe. In space, electronic devices are subject to radiation environment, and 3D PLUS Radiation Assurance Policy has been initially set-up for its Radiation Tolerant Products to cover the effects induced by the space radiation environment:

TOTAL IONIZING DOSE (TID)

Ionization induced in semiconductor materials or associated insulators, such as silicon dioxide layers, can lead to charge trapping or the formation of interface states at the semiconductor-insulator boundary, affecting component behavior or material properties. In MOS devices, the trapped charge can lead to a shift in the gate threshold voltage, and for semiconductors in general, interface states can significantly increase device leakage currents. The device tolerance depends on the semiconductor technology, process, diffusion mask, and wafer fabrication facility. The device tolerance is expressed in krad(Si) based on wafer diffusion lot - batch code.

SINGLE EVENT EFFECTS (SEE)

Effect caused either by direct ionization from a single travelling particle or by recoiling nuclei emitted from a nuclear interaction. The device tolerance is dependent on the semiconductor technology, process, and diffusion mask. The device tolerance is expressed in Linear Energy Transfer (LET) (MeV.cm²/mg) and cross section (cm²/device or cm²/bit) based on die mask revision. The SEE can be distinguished in three main categories:

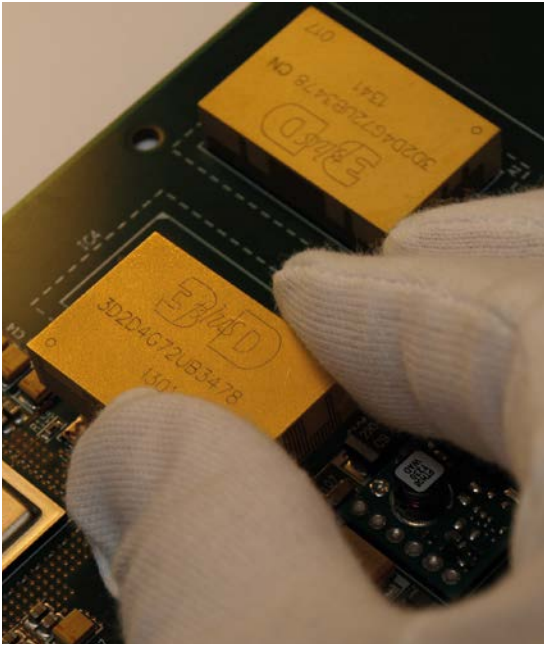
- **Single Event Latch-up (SEL)**
Potentially destructive triggering of a parasitic PNP thyristor structure in a device.
- **Single Event Functional Interrupt (SEFI)**
Interrupt caused by a single particle strike which leads to a temporary nonfunctionality (or interruption of normal operation) of the affected device.
- **Single Event Upset (SEU)**
Single bit flip in a digital element that has been caused either by direct ionisation from a traversing article or by a recoiling nuclei emitted from a nuclear interaction.

CATALOG PRODUCTS

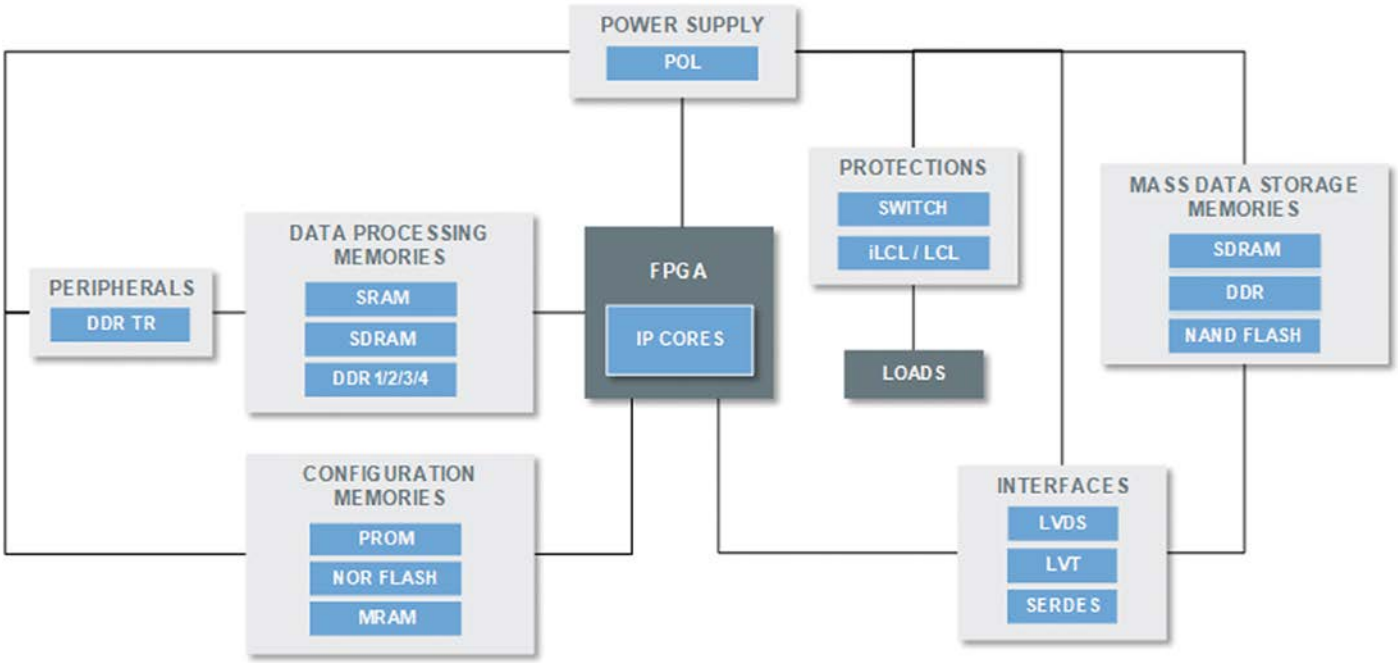
3D PLUS standard product portfolio includes a large range of volatile and non volatile memories, memory Controllers, Computer Cores, Camera Heads, Interfaces, Point-of-Loads, Latch-Up Protections and Termination Regulators.

3D PLUS products are fully space qualified and offer key advantages to our customers' electronic design:

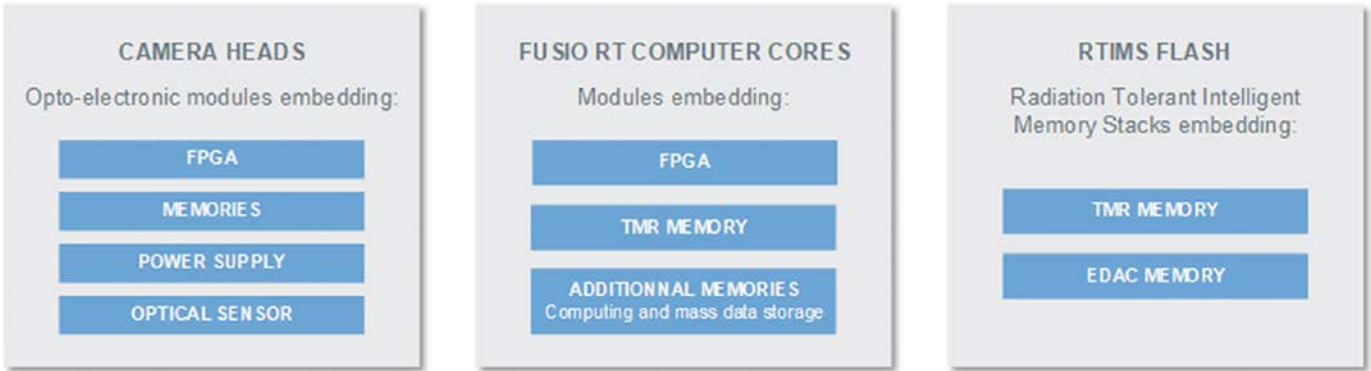
- High reliability
- Excellent signal integrity
- High speed performance
- Radiation hardened by design
- Small form factor
- Flight proven solutions



COMPONENTS



MICRO-SYSTEMS



CUSTOM SOLUTIONS: SYSTEM-IN-PACKAGE

A System-In-Package consists of a number of dissimilar integrated circuits enclosed in a single highly miniaturized package. The SiP performs all or most of the functions of an electronic system, and, it can contain several silicon components (bare die or package) and passive components.

Thanks to the 3D PLUS stacking technology, from a customer's schematic, the generated SiP allows gaining a factor of at least 10 times on weight and volume of the customer PCB. Our strong expertise in Digital, Analog and IP cores combined with our unique know-how in stacking technology for Space Radiation environment make our SiP your missions enablers. 3D PLUS provides a one-stop source for customer's concept analysis, feasibility study, design, manufacturing and test of high reliability and high performance SiP.

1

INITIAL CONCEPT ANALYSIS AND FEASIBILITY STUDY

- Initial requirements analysis
- Initial feasibility study and SiP technology selection
- Manufacturing line selection

2

REQUIREMENTS DETAILED ANALYSIS

- Detailed requirements specification analysis
- Final design trade-offs & feasibility studies
- Product Requirement Specification

3

PRELIMINARY DESIGN

- Functional and physical partitioning analysis
- Module physical architecture and layers' definition
- Preliminary mechanical outline
- Electrical analysis (power, critical signal paths, I/Os,...)
- Thermal and mechanical analysis
- Environmental compatibility analysis including space radiations

4

DETAILED DESIGN

- Detailed design and lay-out of the module
- Electrical Analysis and simulation
- Final thermal and mechanical analysis
- Detailed design report
- Test equipment and procedure definition
- Product validation and qualification plan

5

PROTOTYPES MANUFACTURING AND TEST

- Design and manufacturing of the test specific tools
- Parts and materials procurement
- Manufacturing and test of prototypes
- Validation
- Final updated Product specification and data sheet

6

INDUSTRIALIZATION AND FLIGHT MODELS MANUFACTURING

- Procurement of the additional specific manufacturing and test tools
- Flight Models Manufacturing
- Qualification

VOLATILE MEMORIES

DDR4	DENSITY	CONFIG.	VOLTAGE (V)	ACCESS / CLOCK	RADIATION TOLERANCE				PACKAGE	SCD#
					TiD ⁽¹⁾ krad (Si)	SEL ⁽¹⁾ MeV.cm²/mg	SEU MeV.cm²/mg	SEU Xsection ⁽²⁾ cm²/bit		
3D4D16G72LB2832	16 Gbit	256M x 72	1.2	1200 MHz	100	62.8	2.6	4.10E-10 ⁽³⁾	BGA 259	3DPA-8252
3D4D24G48LB2833	24 Gbit	512M x 48	1.2	1200 MHz	100	62.8	2.6	4.10E-10 ⁽³⁾	BGA 259	3DPA-8256
3D4D32G16LB4827	32 Gbit	2G x 16	1.2	1200 MHz	100	62.8	2.6	4.10E-10 ⁽³⁾⁽⁴⁾	BGA 95	3DFP-0827
3D4D32G72LB2805	32 Gbit	512M x 72	1.2	1200 MHz	100	62.8	2.6	4.10E-10 ⁽³⁾⁽⁴⁾	BGA 259	3DFP-0805
3D4D48G48LB2825	48 Gbit	1G x 48	1.2	1200 MHz	100	62.8	2.6	4.10E-10 ⁽³⁾⁽⁴⁾	BGA 259	3DFP-0825

Temperature: C, I, S. Specific temperature range: -40 °C to 105 °C

DDR3	DENSITY	CONFIG.	VOLTAGE (V)	ACCESS / CLOCK	RADIATION TOLERANCE				PACKAGE	SCD#
					TiD ⁽¹⁾ krad (Si)	SEL ⁽¹⁾ MeV.cm²/mg	SEU MeV.cm²/mg	SEU Xsection ⁽²⁾ cm²/bit		
3D3D16G16YB4751	16 Gbit	1G x 16	1.35 or 1.5	667 MHz	75	67	0.4	1.05E-11	BGA 95	3DPA-7970
3D3D16G72WB2723	16 Gbit	256M x 72	1.35 or 1.5	667 MHz	75	67	0.4	1.05E-11	BGA 199	3DPA-7410
3D3D24G48YB2732	24 Gbit	512M x 48	1.35 or 1.5	667 MHz	75	67	0.4	1.05E-11	BGA 259	3DPA-8141

Temperature: C, I, S. Specific temperature range: -40 °C to 105 °C

DDR2	DENSITY	CONFIG.	VOLTAGE (V)	ACCESS / CLOCK	RADIATION TOLERANCE				PACKAGE	SCD#
					TiD ⁽¹⁾ krad (Si)	SEL ⁽¹⁾ MeV.cm²/mg	SEU MeV.cm²/mg	SEU Xsection ⁽²⁾ cm²/bit		
3D2D1G08US1285	1 Gbit	128M x 8	1.8	125-333 MHz	100	68	2.3	5.6E-11	SOP 74	3DPA-6050
3D2D2G08US2662	2 Gbit	256M x 8	1.8	125-333 MHz	100	68	2.3	5.6E-11	SOP 74	3DPA-6380
3D2D2G16UB2684	2 Gbit	128M x 16	1.8	125-333 MHz	100	68	2.3	5.6E-11	BGA 95	3DPA-6270
3D2D4G08US4661	4 Gbit	512M x 8	1.8	125-333 MHz	100	68	2.3	5.6E-11	SOP 74	3DPA-6090
3D2D4G72UB3652	4 Gbit	64M x 72	1.8	125-333 MHz	100	68	2.3	5.6E-11	BGA 191	3DPA-5600
3D2D6G48UB3687	6 Gbit	128M x 48	1.8	125-333 MHz	100	68	2.3	5.6E-11	BGA 143	3DPA-6630
3D2D6G48UQ3694	6 Gbit	128M x 48	1.8	125-333 MHz	100	68	2.3	5.6E-11	QFP 144	3DPA-6690
3D2D8G08US8663	8 Gbit	1G x 8	1.8	125-200 MHz	100	68	2.3	5.6E-11	SOP 88	3DPA-6100

Temperature: C, I, S. Specific temperature range: -40 °C to 105 °C

DDR1	DENSITY	CONFIG.	VOLTAGE (V)	ACCESS / CLOCK	RADIATION TOLERANCE				PACKAGE	SCD#
					TiD ⁽¹⁾ krad (Si)	SEL ⁽¹⁾ MeV.cm²/mg	SEU MeV.cm²/mg	SEU Xsection ⁽²⁾ cm²/bit		
3D1D2G16TS2174	2 Gbit	128M x 16	2.5	100-200 MHz	50	80	0.7	1E-9	SOP 66	3DPA-3120
3D1D2G32TS2491	2 Gbit	64M x 32	2.5	100-200 MHz	50	80	0.7	1E-9	SOP 86	3DPA-4690
3D1D4G72TB2729	4 Gbit	64M x 72	2.5	100-200 MHz	50	80	0.7	1E-9	BGA 239	3DPA-6150
3D1D8G16TS8466	8 Gbit	512M x 16	2.5	100-200 MHz	50	80	0.7	1E-9	SOP 78	3DPA-4330

Temperature: C, I, S. Specific temperature range: -40 °C to 105 °C

IP CORES	P/N	DESCRIPTION	TARGET FPGAs
RIMC DDR2	3DIPMC0700	Configurable high performing DDR controller mitigating radiation effects to increase memories protection. Memories SEU and SEFI enhanced protection (ECC, scrubbing user definable).	Xilinx (V5QV, Series 7), Microchip (RTG4)
RIMC DDR3	3DIPMC0744		Xilinx (series 7, Ultrascale), Microchip (Polarfire)
RIMC DDR4	3DIPMC0815		Xilinx (Ultrascale), Microchip (Polarfire)
Reed Solomon	3DIPEC0725 3DIPEC0726	Reed Solomon Encoding and Decoding IP Core to provide protection for data storage in space application.	-
			-

(1) Minimum values.

(2) SEU saturated cross section.

(3) Including SBUs from SEFI.

(4) Under evaluation.

SDRAM	DENSITY	CONFIG.	VOLTAGE (V)	ACCESS / CLOCK	RADIATION TOLERANCE				PACKAGE	SCD#
					TiD ⁽¹⁾ krad (Si)	SEL ⁽¹⁾ MeV.cm²/mg	SEU MeV.cm²/mg	SEU Xsection ⁽²⁾ cm²/bit		
3DSD512M16VS1605	512 Mbit	32M x 16	3.3	100-133 MHz	50	80	7	2E-10	SOP 54	3DPA-4780
3DSD1G08VS2685	1 Gbit	128M x 8	3.3	100-133 MHz	50	80	7	2E-10	SOP 54	3DPA-7160
3DSD1G16VS2494	1 Gbit	64M x 16	3.3	100-133 MHz	50	80	7	2E-10	SOP 58	3DPA-4950
3DSD1G32VS2490	1 Gbit	32M x 32	3.3	100-133 MHz	50	80	7	2E-10	SOP 70	3DPA-4840
3DSD2G08VS4686	2 Gbit	256M x 8	3.3	100-133 MHz	50	80	7	2E-10	SOP 54	3DPA-6440
3DSD2G16VS4767	2 Gbit	128M x 16	3.3	100-133 MHz	50	80	7	2E-10	SOP 54	3DPA-7900
3DSD2G32VS4484	2 Gbit	64M x 32	3.3	100-133 MHz	50	80	7	2E-10	SOP 70	3DPA-4660
3DSD2G64VB4488	2 Gbit	32M x 64	3.3	100-133 MHz	50	80	7	2E-10	BGA 119	3DPA-5000
3DSD2G40VS5493	2.56 Gbit	64M x 40	3.3	100-133 MHz	50	80	7	2E-10	SOP 70	3DPA-4760
3DSD3G48VQ6486	3 Gbit	64M x 48	3.3	100-133 MHz	50	80	7	2E-10	QFP 114	3DPA-4670
3DSD4G08VS8613	4 Gbit	512M x 8	3.3	100-133 MHz	50	80	7	2E-10	SOP 58	3DPA-5020
3DSD4G16VS8483	4 Gbit	256M x 16	3.3	100-133 MHz	50	80	7	2E-10	SOP 62	3DPA-4630

Temperature: C, I, S. Specific temperature range: -40 °C to 105 °C

SRAM	DENSITY	CONFIG.	VOLTAGE (V)	ACCESS / CLOCK	RADIATION TOLERANCE				PACKAGE	SCD#
					TiD ⁽¹⁾ krad (Si)	SEL ⁽¹⁾ MeV.cm²/mg	SEU MeV.cm²/mg	SEU Xsection ⁽²⁾ cm²/bit		
3DSR4M08VS1264	4 Mbit	512k x 8	3.3	12 ns	50	80	2	1E-8	SOP 44	3DPA-3350
3DSR4M08CS1647	4 Mbit	512k x 8	5	12 ns	100	110	0.7	6E-8	SOP 44	3DPA-5610
3DSR8M16VS2505	8 Mbit	512k x 16	3.3	12 ns	100	110	0.7	6E-8	SOP 54	3DPA-3300
3DSR8M16CS2510	8 Mbit	512k x 16	5	12 ns	100	110	0.7	6E-8	SOP 54	3DPA-4080
3DSR8M32VS2503	8 Mbit	256k x 32	3.3	12 ns	100	110	0.7	6E-8	SOP 64	3DPA-3290
3DSR16M08CS4660	16 Mbit	2M x 8	5	12 ns	100	110	0.7	6E-8	SOP 44	3DPA-7280
3DSR16M16VS4502	16 Mbit	1M x 16	3.3	12 ns	100	110	0.7	6E-8	SOP 54	3DPA-3280
3DSR16M16CS4512	16 Mbit	1M x 16	5	12 ns	100	110	0.7	6E-8	SOP 54	3DPA-3770
3DSR16M32VS4500	16 Mbit	512k x 32	3.3	12ns	100	110	0.7	6E-8	SOP 64	3DPA-3230
3DSR16M32CS4511	16 Mbit	512k x 32	5	12 ns	100	110	0.7	6E-8	SOP 64	3DPA-3780
3DSR20M40VS2708	20 Mbit	512k x 40	3.3	12 ns	100	110	0.7	6E-8	SOP 84	3DPA-6860
3DSR32M32VS8504	32 Mbit	1M x 32	3.3	12 ns	100	110	0.7	6E-8	SOP 68	3DPA-3800

Temperature: C, I, M

TEMPERATURE RANGES

C: Commercial (0 °C to +70 °C)
I: Industrial (-40 °C to +85 °C)
M: Military (-55 °C to +125 °C)
S: Product specific temperature range

QUALITY GRADES

N: Commercial

B: Industrial

S: Space

ORDERING INFORMATION

Screening level Options
Part Number XX X00X
Temperature range
<http://www.3d-plus.com/how-to-buy.php>

NON VOLATILE MEMORIES

NAND FLASH	DENSITY	CONFIG.	VOLTAGE (V)	ACCESS / CLOCK	RADIATION TOLERANCE				PACKAGE	SCD#
					TID ⁽¹⁾ krad (Si)	SEL ⁽¹⁾ MeV.cm²/mg	SEU MeV.cm²/mg	SEU Xsection ⁽²⁾ cm²/bit		
3DFN8G08VS1706	8 Gbit	1G x 8	3.3	25 ns	60	62.5	1.3	2E-10	SOP 50	3DPA-6760
3DFN16G08VS1712	16 Gbit	2G x 8	3.3	25 ns	60	62.5	1.3	2E-10	SOP 50	3DPA-6750
3DFN16G08VS2705	16 Gbit	2G x 8	3.3	25 ns	60	62.5	1.3	2E-10	SOP 50	3DPA-6770
3DFN32G08VS2711	32 Gbit	4G x 8	3.3	25 ns	60	62.5	1.3	2E-10	SOP 50	3DPA-6740
3DFN32G08VS4704	32 Gbit	4G x 8	3.3	25 ns	60	62.5	1.3	2E-10	SOP 50	3DPA-6780
3DFN64G08VS4309	64 Gbit	8G x 8	3.3	25 ns	60	62.5	1.3	2E-10	SOP 50	3DPA-6730
3DFN64G08VS8695	64 Gbit	8G x 8	3.3	25 ns	60	62.5	1.3	2E-10	SOP 50	3DPA-6790
3DFN64G16VS4679	64 Gbit	4G x 16	3.3	25 ns	60	62.5	1.3	2E-10	SOP 58	3DPA-6980
3DFN64G16VS8710	64 Gbit	4G x 16	3.3	25 ns	60	62.5	1.3	2E-10	SOP 58	3DPA-6800
3DFN128G08VS8308	128 Gbit	16G x 8	3.3	25 ns	60	62.5	1.3	2E-10	SOP 50	3DPA-6720
3DFN128G16VS8709	128 Gbit	8G x 16	3.3	25 ns	60	62.5	1.3	2E-10	SOP 50	3DPA-6990
3DFN128G16VS8810	128 Gbit	8G x 16	3.3	25 ns	60	62.5	1.3	2E-10	SOP 50	3DPA-8198
3DFN128G08US8761	128 Gbit	16G x 8	1.8 and 3.3	12 ns	60	60	1.3	2E-10	SOP 54	3DPA-7920

Temperature: C, I, M

NOR FLASH	DENSITY	CONFIG.	VOLTAGE (V)	ACCESS / CLOCK	RADIATION TOLERANCE				PACKAGE	SCD#
					TID ⁽¹⁾ krad (Si)	SEL ⁽¹⁾ MeV.cm²/mg	SEU MeV.cm²/mg	SEU Xsection ⁽²⁾ cm²/bit		
3DFO64M16VS1281	64 Mbit	4M x 16	3.0	90 ns	20 ⁽³⁾	51.2	10	1E-11	SOP 54	3DPA-2710
3DFO128M16VS2282	128 Mbit	8M x 16	3.0	90 ns	20 ⁽³⁾	51.2	10	1E-11	SOP 54	3DPA-2760
3DFO256M16VS4105	256 Mbit	16M x 16	3.0	90 ns	20 ⁽³⁾	51.2	10	1E-11	SOP 54	3DPA-2920
3DFO256M16VS4269	256 Mbit	16M x 16	3.0	90 ns	20 ⁽³⁾	51.2	10	1E-11	SOP 54	3DPA-2330
3DFO512M16VS8492	512 Mbit	32M x 16	3.0	90 ns	20 ⁽³⁾	51.2	10	1E-11	SOP 56	3DPA-5050
3DFO2G16VS4214	2 Gbit	128M x 16	3.3	90 ns	20 ⁽³⁾	> 60	1.8	8E-11	SOP 60	3DPA-5200
3DFO2G08VS4215	2 Gbit	256M x 8	3.3	90 ns	20 ⁽³⁾	> 60	1.8	8E-11	SOP 60	3DPA-3620

Temperature: C, I, M

SPI NOR FLASH	DENSITY	CONFIG.	VOLTAGE (V)	ACCESS / CLOCK	RADIATION TOLERANCE				PACKAGE	SCD#
					TID ⁽¹⁾ krad (Si)	SEL ⁽¹⁾ MeV.cm²/mg	SEU MeV.cm²/mg	SEU Xsection ⁽²⁾ cm²/bit		
3DFS256M04VS1740	256 Mbit	SPI/QSPI	3.0	50-133 MHz	20 ⁽³⁾	62.5	15	7.5E-11	SOP 18	3DPA-7590
3DFS512M04VS2722	512 Mbit	SPI/QSPI	3.0	50-133 MHz	20 ⁽³⁾	62.5	15	7.5E-11	SOP 18	3DPA-7450

Temperature: C, I, S. Specific temperature range: -55 °C to 105 °C.

TMR	DENSITY	CONFIG.	VOLTAGE (V)	ACCESS / CLOCK	RADIATION TOLERANCE			TEMP.	PACKAGE	SCD#
					TID ⁽¹⁾ krad (Si)	SEL ⁽¹⁾ MeV.cm²/mg	SEU MeV.cm²/mg			
3DFS128M01VS2728	128 Mbit	TMR SPI	3.3	40 MHz	40	62.5	Immune ⁽⁴⁾	C, I, M	SOP 20	3DPA-7650
3DFS256M04VS2801	256 Mbit	TMR SPI	3.3	50 MHz	40	62.5	Immune ⁽⁴⁾	C, I, S	SOP 24	3DPA-8204

Specific temperature range: -55°C to +105°C

(1) Minimum values.
(2) SEU saturated cross section.
(3) TID value under the condition of 50% duty cycle.

(4) Immune by Radiation Intelligent TMR design.
(5) Please refer to Application Note 3300-6674

RTIMS FLASH	DENSITY	CONFIG.	VOLTAGE (V)	ACCESS / CLOCK	RADIATION TOLERANCE				PACKAGE	SCD#
					TID ⁽¹⁾ krad (Si)	SEL ⁽¹⁾ MeV.cm²/mg	SEU MeV.cm²/mg	SEU Xsection ⁽²⁾ cm²/bit		
3DSS24G08VS3626	24 Gbit	3G x 8	3.3	25 ns	50	60	Immune ⁽⁴⁾	-	SOP 38	3DPA-5780

Temperature: C, I

PROM	DENSITY	CONFIG.	VOLTAGE (V)	ACCESS / CLOCK	RADIATION TOLERANCE				PACKAGE	SCD#
					TID ⁽¹⁾ krad (Si)	SEL ⁽¹⁾ MeV.cm²/mg	SEU MeV.cm²/mg	SEU Xsection ⁽²⁾ cm²/bit		
3DPO32M08VS1419	32 Mbit	4M x 8	3.3	33 MHz	50	120	Immune	-	SOP 44	3DPA-5080
3DPO64M08VS2299	64 Mbit	8M x 8	3.3	33 MHz	50	120	Immune	-	SOP 44	3DPA-3450
3DPO128M08VS4667	128 Mbit	16M x 8	3.3	33 MHz	50	120	Immune	-	SOP 44	3DPA-5960

Temperature: C, I, M

MRAM	DENSITY	CONFIG.	VOLTAGE (V)	ACCESS / CLOCK	RADIATION TOLERANCE				PACKAGE	SCD#
					TID ⁽¹⁾ krad (Si)	SEL ⁽¹⁾ MeV.cm²/mg	SEU MeV.cm²/mg	SEU Xsection ⁽²⁾ cm²/bit		
3DMR1M08VS1426	1 Mbit	128k x 8	3.3	40 ns	50	85	Immune	-	SOP 44	3DPA-4730
3DMR2M16VS2427	2 Mbit	128k x 16	3.3	40 ns	50	85	Immune	-	SOP 54	3DPA-4650
3DMR4M08VS4428	4 Mbit	512k x 8	3.3	40 ns	50	85	Immune	-	SOP 44	3DPA-4520
3DMR8M08VS8666	8 Mbit	1M x 8	3.3	40 ns	50	85	Immune	-	SOP 44	3DPA-6670
3DMR8M32VS8420	8 Mbit	256k x 32	3.3	40 ns	50	85	Immune	-	SOP 68	3DPA-4100
3DMR10M40VS5688	10 Mbit	256k x 40	3.3	40 ns	50	85	Immune	-	SOP 96	3DPA-6460
3DMR64M08VS4476	64 Mbit	8M x 8	3.3	40 ns	50	85 ⁽⁵⁾	Immune	-	SOP 54	3DPA-6170

Temperature: C, I, S. Specific temperature range: -55 °C to +105 °C

TEMPERATURE RANGES

C: Commercial (0 °C to +70 °C)
I: Industrial (-40 °C to +85 °C)
M: Military (-55 °C to +125 °C)
S: Product specific temperature range

QUALITY GRADES

N: Commercial
B: Industrial
S: Space

ORDERING INFORMATION

Screening level Options
Part Number XX X00X
Temperature range
<http://www.3d-plus.com/how-to-buy.php>

DDR4 ECOSYSTEM

FULLY QUALIFIED DDR SOLUTION

OVERVIEW

3D PLUS DDR4 memory modules are part of a complete DDR4 ecosystem, providing fully space qualified memory subsystem solutions and operating in a temperature range from -40°C to +105°C.

3D PLUS's DDR4 Ecosystem includes:

- A wide range of **DDR4 memories** to handle high speed computing data requirements;
- A **DDR4 Termination Regulator** to efficiently provide stable V_{TT} to DDR4 components;
- **DDR4 Radiation Intelligent Memory Controller (RIMC)** IP that allows enhanced protection and mitigation against SEUs/SEFIs.

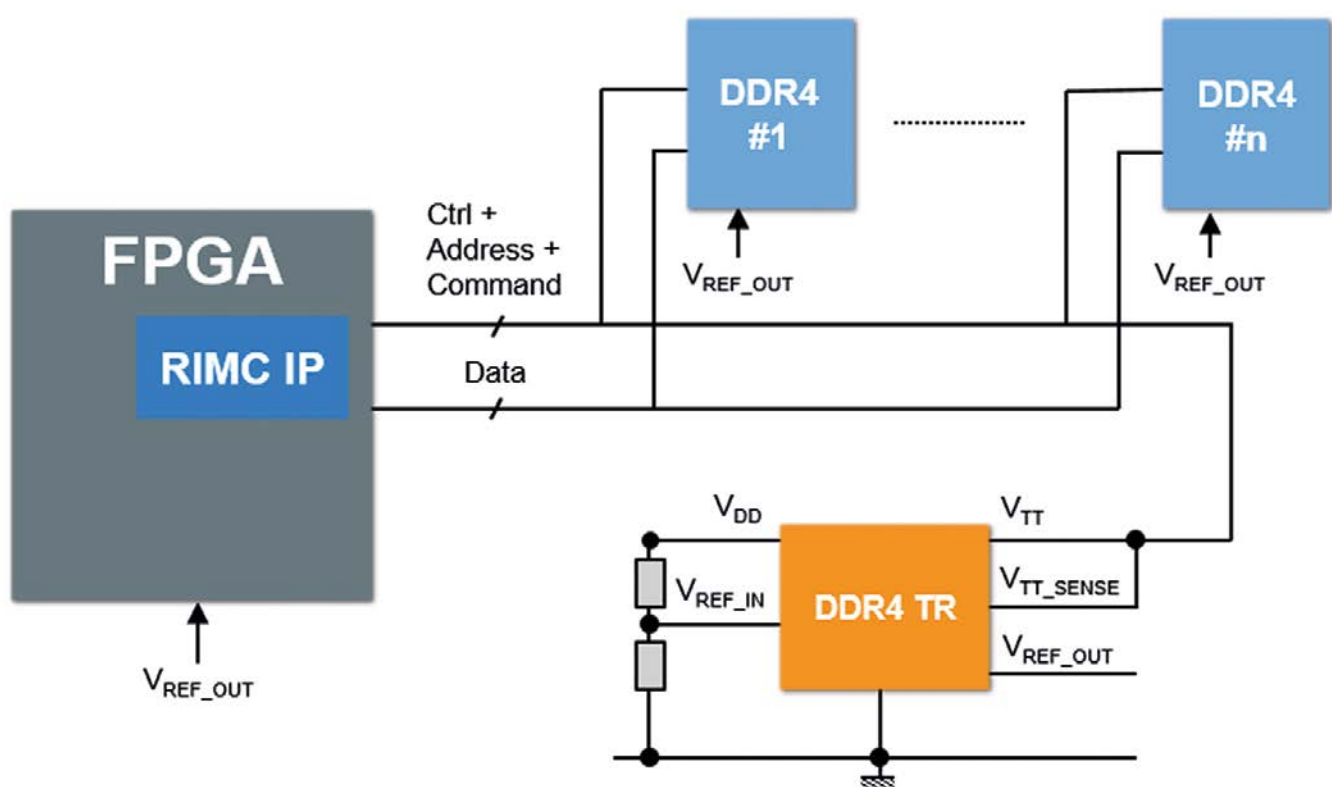
APPLICATIONS

3D PLUS DDR4 products are the perfect fit for systems requiring best-in class, high performance memories, qualified for use in Space applications.

These memories have been designed to be used with space grade microprocessors, DSPs and FPGAs.

KEY BENEFITS

- Long lifecycle products with proven reliability in Space
- Ruggedized, fully validated complete DDR4 solution
- Significant board area savings
- Optimized development time and costs



DDR4 SDRAM MEMORY

KEY FEATURES

- Radiation tolerant and highly reliable DDR4
- High frequency: $f = 1.2$ GHz
- High density: up to 48 Gbit
- Large Data Bus width: x72 bits (with 8 bits ECC), x48 bits, x16 bits



RADIATION DATA

- TID > 100 krad(Si)
- SEL immune up to LET > 62.8 MeV.cm²/mg
- SEU/SEFI protection / mitigation thanks to RIMC IP

RADIATION INTELLIGENT MEMORY CONTROLLER (RIMC)

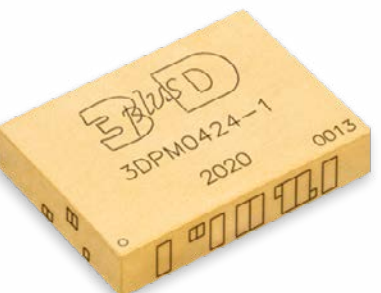
KEY FEATURES

- High performance DDR4 controller
- Scrubbing mechanisms (user-definable)
- Configurable user interface AMBA compliant AXI/APB
- Enhanced SEU protection with ECC (Hamming or Reed Solomon)
- SEFI mitigation

DDR4 TERMINATION REGULATOR

KEY FEATURES

- Source and Sink current capability $\pm 2A$
- JESD79-4C compliant solution
- Thermal protection
- Eliminates the need for external capacitors



SIGNAL INTEGRITY AND ELECTRICAL PERFORMANCES

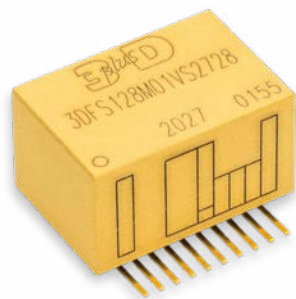
- Impedance Adaptation: Due to DDR4 high data transmission speeds, termination is one of the major challenges when using DDR4 memories.
- Signal Integrity: Address, Control and Command lines have to be terminated through V_{TT} termination voltage to avoid noise, crosstalk and timing degradation during data accesses.
- Minimizes Power Dissipation: One can achieve lower power dissipation with an integrated TR as opposed to a discrete solution.

RADIATION DATA

- TID > 75 krad(Si)
- SEL immune up to LET > 62.5 MeV.cm²/mg

TMR NOR FLASH MEMORIES

3DFS128M01VS2728
3DFS256M04VS2801



PRODUCT OVERVIEW

The TMR SPI NOR Flash family offers modules embedding TMR (Triple Modular Redundancy) and power switch. The memories operating with an SPI interface guarantees data retention and high level of radiation performances. Used as configuration memories, they preserve data integrity from the start until the end of the mission in harsh environments.

KEY FEATURES

- Radiation Hardened By Design (RHBD)
- Low power with wide temperature range
- Single 3.3 V voltage supply
- Triple Modular Redundancy (TMR) embedded
- Data retention > 20 years
- Small footprint and pin count

RADIATION DATA

- TID > 40 krad(Si) in OFF mode
- SEL LET > 62.5 MeV.cm²/mg
- SEU immune (thanks to TMR)
- SET and SEFI mitigated by design

KEY BENEFITS

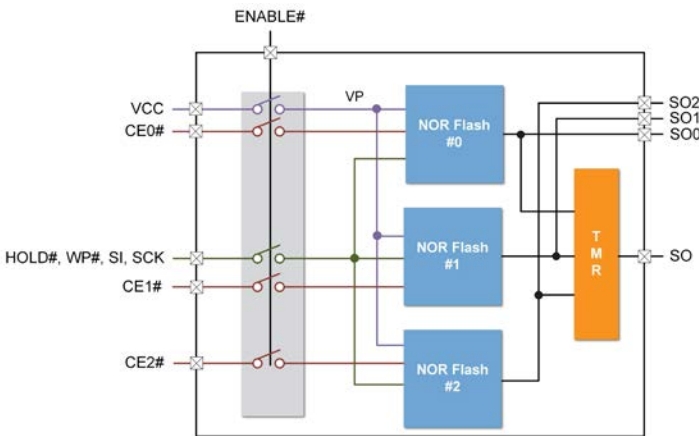
- Highly reliable space grade solution
- Ensure data integrity in harsh environment
- Area saving
- Plug and play solution
- Intelligent 256 Mbit memory:
 - Easy and secured in-flight reconfigurable
 - Error Management (ECC_Flag)

APPLICATIONS

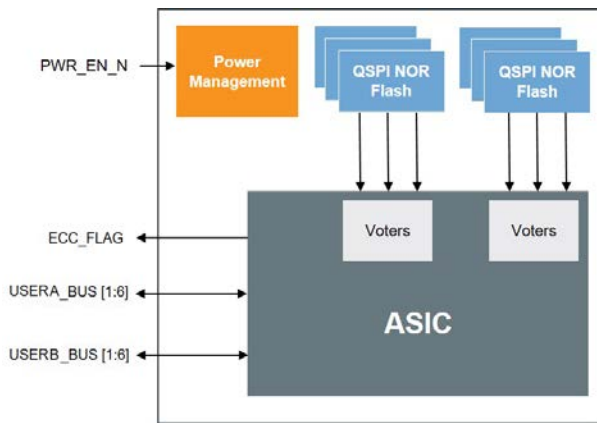
- Boot code storage for microcontrollers and microprocessors
- FPGA configuration bitstream storage

PART NUMBER	DENSITY	MODE	ACCESS/CLOCK	OPERATING TEMPERATURE	PACKAGE
3DFS128M01VS2728	128 Mbit	SPI	40 MHz	-55 °C to +125 °C	SOP 20
3DFS256M04VS2801	256 Mbit	SPI, QSPI, dual QSPI	50 MHz	-55 °C to +105 °C	SOP 24

3DFS128M01VS2728 BLOCK DIAGRAM



3DFS256M04VS2801 BLOCK DIAGRAM



MASS DATA STORAGE MEMORIES

RADIATION TOLERANT AND INTELLIGENT MEMORY STACKS

3DSS24G08VS3626



PRODUCT OVERVIEW

The NAND Flash Radiation Tolerant and Intelligent Memory Stack (RTIMS FLASH) is a user-friendly, plug-and-play, high density module regarding its embedded SEU, SEL and SEFI protections.

KEY BENEFITS

- Radiation protected & hardened by design NAND Flash memory,
- Error free guaranteed sector 0 for boot data storage,
- Embedded format, fetch failing block, sector logic generation commands for initialization,
- Internal registers for configuration, operation status and telemetry operation.

RADIATION DATA

- TID > 50 krad(Si)
- SEL > 60 MeV.cm²/mg
- SEU (TMR Mode, LEO) < 9.4 x 10⁻²⁶ error/bit/year
- SEU (EDAC Mode, LEO) < 1.7 x 10⁻²⁵ error/bit/year
- HC SEFI Immune by design

KEY FEATURES

- Single 3.3 V Power Supply
- Up to 24 Gbit of Space qualified NAND Flash memory
- Bad block free - continuous logic sectors
- Access time: 99 Mbit/s (write), 287 Mbit/s (read)
- Data retention: 10 years
- Operating temperature range: -40 °C to +85 °C
- Package: SOP 38 - pitch: 1.27 mm
- Dimensions: 31 mm x 28 mm x 11.2 mm
- Weight: 16 g

SYNCHRONOUS NAND FLASH

3DFN128G08US8761



PRODUCT OVERVIEW

The 3DFN128G08US8761 is a high-density non-volatile NAND FLASH module operating in both asynchronous and synchronous modes. This module is suited for data recorder (mass data storage).

KEY BENEFITS

- Twice as fast as asynchronous NAND Flash
- Low power consumption
- Low usage of 3.3 V I/O pins on boards

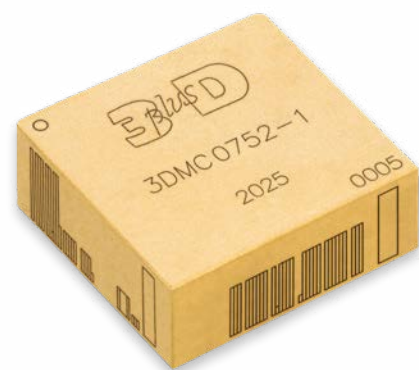
RADIATION DATA

- TID > 60 krad(Si)
- SEL LET > 62.5 MeV.cm²/mg

KEY FEATURES

- ONFI 2.1 compliant
- Dual voltage operation: 3.3 V for core and 1.8 V for I/O
- Synchronous I/O performance
 - Read/write throughput per pin: 166 MT/s
- Hardware Data Protection
- Endurance: 100k Program/Erase Cycles with ECC
- Data Retention: 10 years
- Operating temperature range: -55 °C to 125 °C
- Package: SOP 54 - Pitch: 0.65 mm
- Dimensions: 15.1 mm x 20.6 mm x 12.6 mm
- Weight: 9.1 g

FUSIO RT COMPUTER CORES



PRODUCT OVERVIEW

The FUSIO RT is a complete reconfigurable Computer Core integrating a space grade SRAM-based FPGA and its configuration memory, with additional computing and mass storage memory, in one miniaturized module.

KEY BENEFITS

- Highly integrated Space Computer Core
- European FPGA
- Reduces significantly the board area
- Modular, easy to use
- Proven radiation hardened design
- Well suited for use in high reliability, high performance and high density system applications

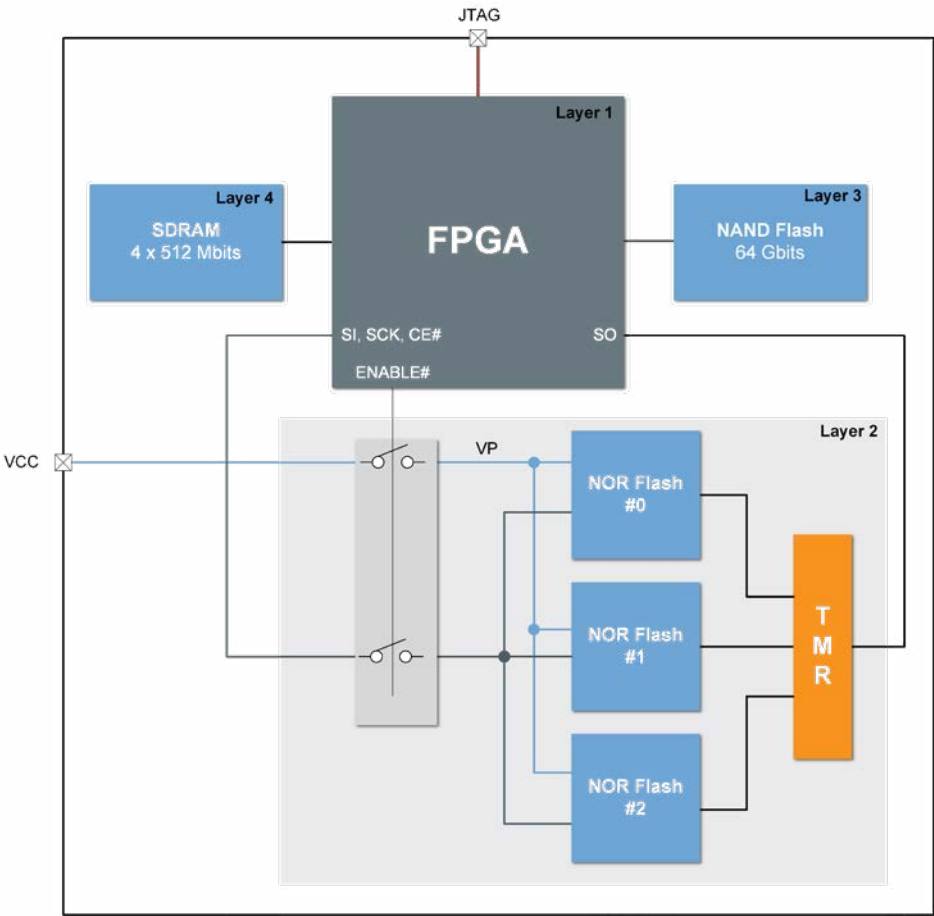
RADIATION DATA

- TID > 40 krad(Si)
- SEL LET > 60 MeV.cm²/mg

KEY FEATURES

- Power Supply: 1.2 V, 2.5 V, 3.3 V
- NX1H35AS: Brave NG_Medium European FPGA
 - 550k ASIC gates
 - Integrated space wire interface
 - 800 Mbit/s I/O support
- FPGA loading at 25 MHz (via internal Flash NOR)
- SPI NOR Characteristics:
 - Triple 128 Mbit / 16 M-Byte
 - Triple Modular Redundancy (TMR) integrated
 - Enhanced TID implementation
- Optional NAND and SDRAM memory integrated into module
- Operating temperature range: -40 °C to +105 °C
- Package: 483 balls BGA, 263 users I/Os
- Dimensions: 32 mm x 32 mm
- Weight: < 25 g

	CONFIGURATION	PART NUMBER	DESCRIPTION
FUSIO RT BASIC CONFIGURATION	2 LAYERS 	3DMC0753	FPGA TMR 128 Mbit SPI NOR Flash (triplicated)
FUSIO RT NAND FLASH	3 LAYERS 	3DMC0754	FPGA TMR 128 Mbit SPI NOR Flash (triplicated) 64 Gbit NAND Flash
FUSIO RT SDRAM	3 LAYERS 	3DMC0755	FPGA TMR 128 Mbit SPI NOR Flash (triplicated) 2 Gbit SDRAM
FUSIO RT FULL OPTIONS	4 LAYERS 	3DMC0752	FPGA TMR 128 Mbit SPI NOR Flash (triplicated) 64 Gbit NAND Flash 2 Gbit SDRAM



MEMORY IP CORES

The FUSIO RT product family comes with dedicated IP Cores for the the memories integrated in the different configurations.

The 3DIPMC0813 RIMC NAND Flash and the 3DIPMCC0816 RIMC SDRAM are Radiation Intelligent Memory Controllers IP Cores designed specifically for the FUSIO RT module to offer a radiation hardened solution of the FUSIO RT configuration using the corresponding memory.

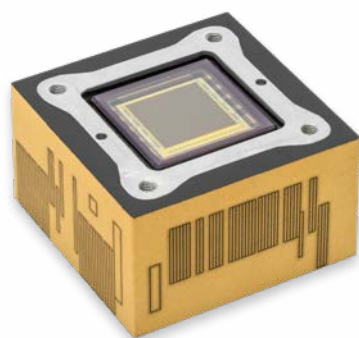
IP CORE	PART NUMBER
NAND FLASH IP CORE	3DIPMC0813
SDRAM IP CORE	3DIPMC0816

EVALUATION BOARD

The 3DEV075x FUSIO RT Development Kit is a ready to use development board integrating a FUSIO RT module (configuration of your choice). The FUSIO RT Development kit allows customers to get familiar with the FUSIO RT developments tools and allows for early prototyping of their application.



CMOS CAMERA HEADS



KEY FEATURES

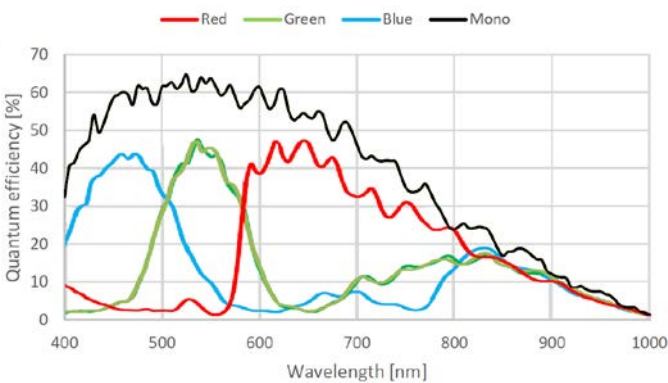
4 Mpx Camera Head:

- 2048x2048 Global Shutter Visible CMOS image sensor (monochrome/RGB)
- Up to 7 frame/s (full frame, 12-bit mode)
- Up to 16 frame/s (full frame, 10-bits mode)
- 1 Gbit SDRAM and 8 Gbit NAND Flash memories
- Read Noise : 13 e-, Dark current : 125 e-/s
- FWC : 10k e
- Operating temperature range: -40 °C to +70 °C
- Dimensions: 35 mm x 35 mm x 23mm
- Weight: 64 g

12 Mpx Camera Head:

- 4096x3000 Global Shutter Visible CMOS image sensor (monochrome/RGB)
- Up to 43 frame/s (full frame, 12-bit mode)
- Up to 68 frame/s (full frame, 10-bits mode)
- 8 Gbit DDR3 SDRAM and 48 Gbit Synchronous NAND Flash memories
- Read Noise : 2 e-, Dark current : 10 e-/s
- FWC : 11k e
- Operating temperature range: -40 °C to +70 °C
- Dimensions: 40 mm x 40 mm x 35 mm
- Weight: 100 g

Spectral response for 4 Mpx Camera Head



PRODUCT OVERVIEW

3D PLUS Camera Heads Family offers high density, high performance and high resolution CMOS camera solutions for space applications. These solutions bring together a high resolution CMOS image sensor with an FPGA-based electronic architecture for control of the sensor, image pre-processing, and interfacing of the camera with a space equipment. This generic camera architecture allows for a wide range of applications, from scientific observation, equipment or payload monitoring, to star-tracking or GNC applications.

KEY BENEFITS

- High degree of integration with an all-in-one solution
- High versatility with user-reconfigurable FPGA-based architecture
- Easy optics centering, focal plans alignment and system integration thanks to dedicated mechanical references
- Optimized interfaces
- Fully qualified for space applications
- More than 16 years of experience with Space camera heads designing

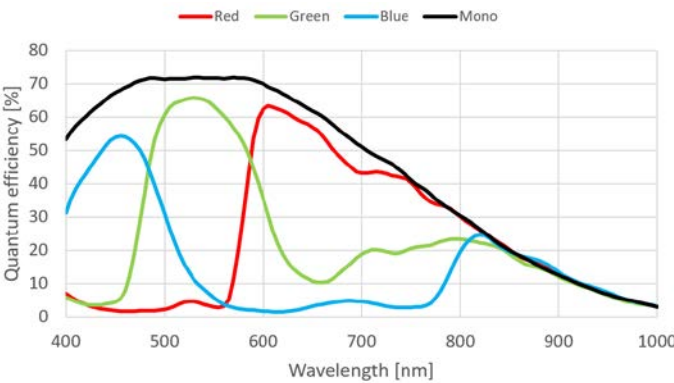
APPLICATIONS

- Earth observation, planetology
- Platform monitoring (satellite, rover)
- GNC applications (tracking, RdV)

RADIATION DATA

- TID > 40 krad(Si)
- SEL LET > 62.5 MeV.cm²/mg

Spectral response for 12 Mpx Camera Head



FIRMWARE ENVIRONMENT

The 3D PLUS space camera heads family offers several firmware options for the configuration of its cameras, all compliant with ECSS-Q-ST-60-02C ESA Standard.

STANDARD FLIGHT CODE

3D PLUS standard flight codes provide a simple and efficient space qualified ready-to-use solution for the 3D PLUS camera heads to support a quick integration in equipment. It allows a simple management of the sensor and of the internal memories of the camera heads and offers a Standard SpaceWire for easy interfacing with space equipments.

SDRAM CONTROLLER

The 3D PLUS SDRAM Controller IP Core allows a simple stream oriented uses of the SDRAM memories of the camera heads and integrates radiation mitigation functionalities for space applications purposes.

NAND FLASH CONTROLLER

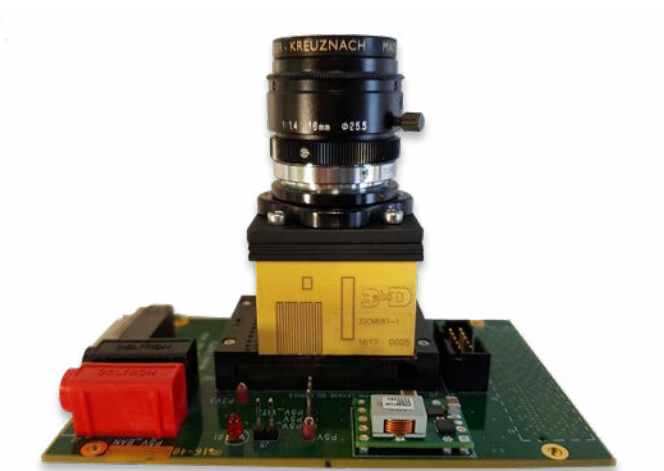
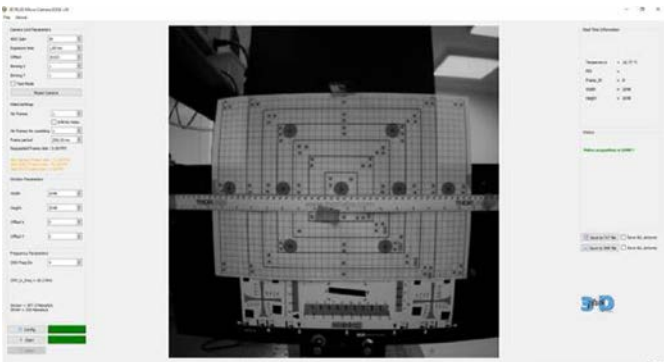
The 3D PLUS NAND Flash Controller IP Cores allow a simple stream oriented use of the NAND Flash camera heads and offers non-volatile storage capabilities to the cameras. They integrate an automatic management of bad blocks of the memory, as well as radiations mitigation functionalities for space application purpose.

SENSOR CONTROLLER

The 3D PLUS CMOS Sensor Controller IP Cores provide a simple control of the CMOS sensors of the Camera heads. They allow the configuration of all sensor's registers as well as several preprocessing functionalities (windowing, binning).

EVALUATION KIT

The camera evaluation kit is a ready to use evaluation environment compatible with 3D PLUS Camera Heads products, and provide a simple software interface to support the use and integration of the camera heads. It allows configuration of the camera heads, and the request of images and video sequences, to be displayed or stored for processing and evaluation.



	P/N	CONFIG.	CONTROLLERS IP CORES				FLIGHT CODE
			SDRAM	DDR3	NAND FLASH	SENSOR	
4 MPX	3DCM734	Monochrome	3DIPCC0735	-	3DIPCC0736	3DIPCC0737	3DIPCC0746
	3DCM739	RGB					
12 MPX	3DCM800	Monochrome	-	3DIPCC0835	3DIPCC0836	3DIPCC0837	3DIPCC0838
	3DCM828	RGB					

INTERFACES

LVDS

- High Speed Performance: > 400 Mbit/s (200 MHz) switching rates
 - Excellent Differential Skew and Propagation Delay
 - Compatible with IEEE 1596.3 SCI and ANSI/TIA/EIA-644 LVDS standards
- Very High Density - small footprint
 - Suitable for SpaceWire data links
 - Cold Sparing Capability
 - Ultra Low Power Dissipation



LVDS	DESCRIPTION	I/Os	VOLTAGE (V)	SWITCHING RATE (Mbit/s)	RADIATION TOLERANCE			PACKAGE	SCD#
					TID krad (Si)	SEL MeV.cm²/mg	SEU MeV.cm²/mg		
3DLV3108VS1372	Octal Driver	LVTTTL / LVCMOS	3.3	> 400	100	> 80	Immune	SOP 34	3DPA-3740
3DLV3208VS1373	Octal Receiver	LVTTTL / LVCMOS	3.3	> 400	100	> 80	Immune	SOP 34	3DPA-3750
3DLV3304VS1374	Quad Driver & Receiver	LVTTTL / LVCMOS	3.3	> 400	100	> 80	Immune	SOP 34	3DPA-3820
3DLV3104VS1617	Quad Driver	LVTTTL / LVCMOS	3.3	> 400	100	> 80	Immune	SOP 16	3DPA-5230
3DLV3204VS1618	Quad Receiver	LVTTTL / LVCMOS	3.3	> 400	100	> 80	Immune	SOP 16	3DPA-5240
3DLV3302VS1619	Dual Driver & Receiver	LVTTTL / LVCMOS	3.3	> 400	100	> 80	Immune	SOP 18	3DPA-5250
3DLV3408VS1715	Octal Repeater	LVTTTL / LVCMOS	3.3	> 400	100	> 80	Immune	SOP 48	3DPA-7020

Temperature: C, I, M

LVT

- Extended Voltage Range
 - High Current Driving Capability
 - High Speed Outputs
 - Bus Hold Provided on Both Sides
- Cold Spare Function
 - Space Quality Grade
 - Compact Size and Low Weight



LVT	DESCRIPTION	VOLTAGE (V)	PROPAGATION DELAY (ns)	RADIATION TOLERANCE			PACKAGE	SCD#
				TID krad (Si)	SEL MeV.cm²/mg	SEU MeV.cm²/mg		
3DLT162245VS1639	Dual 16-bit buffer Transceiver	1.8 - 3.6	3.2	100	Immune	Immune	SOP 82	3DPA-5580
3DLT164245VS1645	Dual 16-bit bidirectional Level Shifter 3.3 V to 5 V	2.7 - 5.5	15	100	Immune	Immune	SOP 82	3DPA-6390
3DLT163245US1696	Dual 16-bit bidirectional Level Shifter 1.8 V to 3.3 V	1.4 - 3.6	5.5	100	Immune	Immune	SOP 82	3DPA-6700

Temperature: C, I, M

SERDES

- Up to 1.428 Gbit/s Data Rate
 - PLL Frequency Range 20 to 68 MHz
 - Low Power Consumption
 - Power-down Mode
 - Compact size and Low weight
- Requires no External Components
 - Rising Clock Edge Trigger
 - Space Qualified Technology
 - Compact size and Low weight



SERDES	CONFIGURATION	VOLTAGE (V)	DATA THROUGHPUT (Gbit/s)	RADIATION TOLERANCE			PACKAGE	SCD#
				TID krad (Si)	SEL MeV.cm²/mg	SEU MeV.cm²/mg		
3DLV21721VS2622	21:3 Serializer	3.3	1.428	> 60	> 67	-	SOP 56	3DPA-5210
3DLV21821VS2623	3:21 Deserializer	3.3	1.428	> 60	> 67	-	SOP 56	3DPA-5220

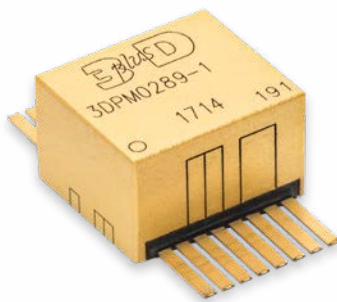
Temperature: C, I, M

TEMPERATURE RANGES
C: Commercial (0 °C to +70 °C)
I: Industrial (-40 °C to +85 °C)
M: Military (-55 °C to +125 °C)

QUALITY GRADES
N: Commercial
B: Industrial
S: Space

ORDERING INFORMATION
Quality Grade — Options
Part Number XX XXX
Temperature Range —
http://www.3d-plus.com/how-to-buy.php

POL CONVERTERS



PRODUCT OVERVIEW

PoL Converter modules provide high performances, high reliability, compact size and low weight for space applications. Based on a Buck topology, they are an excellent solution for low voltage power distribution systems designed around high-speed digital electronics such as ASICs, FPGAs (Microsemi, XILINX,...) and memories (SDRAM, DDR, DDR2, DDR3,...). All input and output filters come integrated within the module. Its output voltage can be adjusted by use of an external resistor.

KEY FEATURES

- Excellent dynamic performances
- High efficiency
- Integrated I/O EMC filters
- Supports Current Sharing

APPLICATIONS

- Low power distributed power architectures
- Power supply for digital loads (FPGA, ASICs, memory banks...)

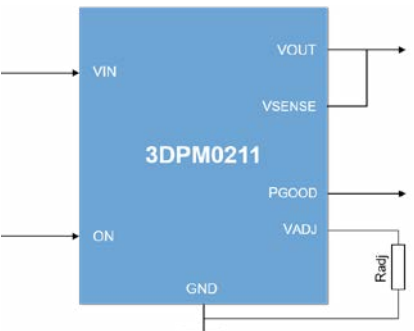
KEY BENEFITS

- All-in-one component - all filters included
- Easy to use - Plug and Play intelligent power solution
- Modular design for distributed power architectures

P/N	Input voltage	Output voltage	Output current	TID krad(Si)	SEL/SET MeV.cm²/mg	Operating Temperature	Package
3DPM0211	5V ±10%	1.2V to 3.8V	5A	50	80	-40 °C to +95 °C	14-Gull-Wing
3DPM0289	4.5V to 12V	1V to 5V	2A	40	60	-40 °C to +85 °C	16-Gull-Wing
3DPM0349	3V to 6V	0.8V to 3.3V	7A	50	60	-40 °C to +105 °C	28-QFP

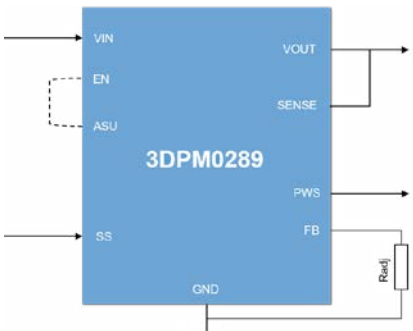
3DPM0211

Dimensions: 26.5 x 25 x 10 mm



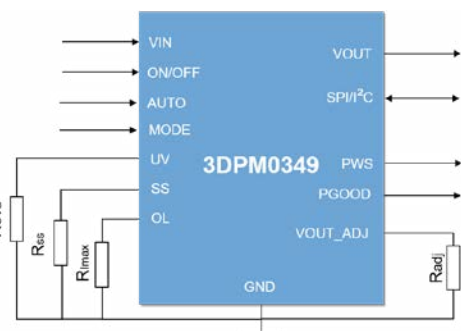
3DPM0289

Dimensions: 12.5 x 11 x 9.4 mm



3DPM0349

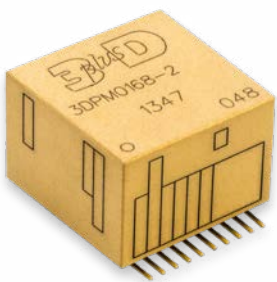
Dimensions: 28 x 28 x 10 mm



PROTECTION ICs

LATCH-UP CURRENT LIMITER

3DPM0168

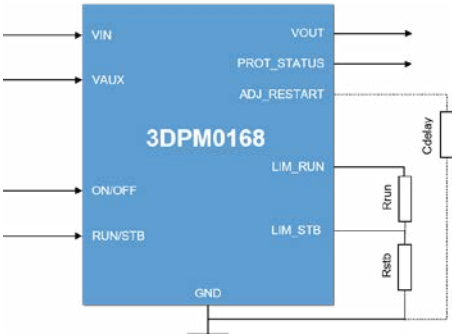


PRODUCT OVERVIEW

Advanced high performance semiconductor devices can be sensitive to Single Event Latch-Up (SEL) effects when exposed to radiation in the space environment. Even though SEL is a very rare event, it can lead to device destruction and therefore must be mitigated to ensure the reliability and success of the mission. The LCL is the best solution for power lines protection.

KEY BENEFITS

- Highly integrated solution
- All-in-one component - two resistors needed
- Generic design to protect any type of load



KEY FEATURES

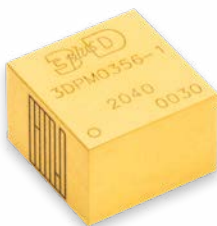
- Latch-up protection from 0.8V to 5.5V and up to 2A
- Very fast switch-off time (10µs max)
- Very low Voltage drop across the LCL
- Adjustable Run & Standby threshold currents
- Automatic reconnection or through ON/OFF command
- Adjustable delay for automatic reconnection
- Protection Status signal for system supervision
- Operating temperature range: -40 °C to +115 °C
- Package: SOP 20 - Pitch: 1.27 mm
- Dimensions: 15 mm x 15 mm x 12 mm
- Weight: 6.5 g

RADIATION DATA

- TID > 50 krad(Si)
- SEL Immune
- SET Immune

SWITCH

3DPM0356

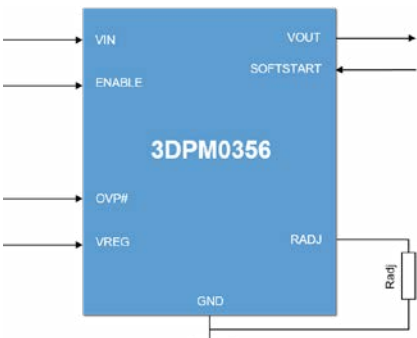


PRODUCT OVERVIEW

The 3DPM0356 switch module, has been designed for isolation purposes. This module is a mono-directional blocking Switch, configured to provide a fast response time against any kind of perturbation. The Switch represents a key element to guarantee a fault tolerant power supply architecture in Space applications.

KEY BENEFITS

- All-in-one component – one external resistor needed
- Simplified design – smallest footprint w. OVC protection included
- Generic design to protect any type of load



KEY FEATURES

- Protection from 3 V to 12 V and up to 7A
- Fast switch-off time (20 µs max for Over-current detection)
- Enable and OVP control inputs
- Current limitation threshold can be set through an external resistor
- Rise Time can be controlled through an external capacitor
- Operating temperature range: -40 °C to +105 °C
- Package: BGA 48 - Pitch: 1.27 mm
- Dimensions: 13 mm x 13 mm x 9.5 mm
- Weight: 4 g

RADIATION DATA

- TID > 50 krad(Si)
- SEL Immune
- SET Immune

DDR TERMINATION REGULATORS

PRODUCT OVERVIEW

One of the major challenges when using DDR memory system architectures is to ensure signal integrity during data accesses. Termination Regulator modules are compact, flexible and Rad Hard By Design. Able to sink and source current while regulating VTT voltage within the limits defined by JEDEC standards. As part of the DDR Ecosystem, they efficiently maintain fast transient response times; offer a remote sensing function; support control via Command pin and provide all power requirements for DDR VTT bus designs.

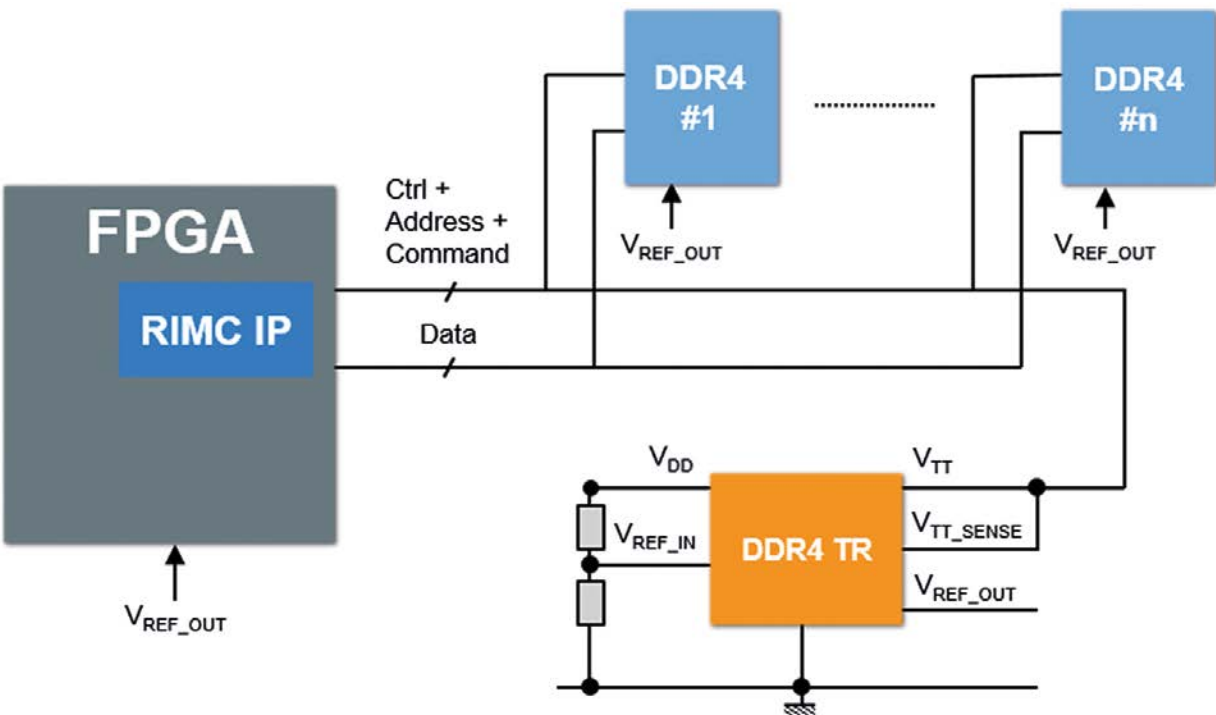
KEY FEATURES

- Source and sink current capabilities up to ±2A
- JEDEC compliant solution
- Thermal protection
- Eliminate the need for external capacitors

KEY BENEFITS

- Global radiation tolerant solution
- Highly integrated – All in one
- Significant board area savings

P/N	DDR Technology	Input voltage	Output current	TID krad(Si)	SEE MeV.cm²/mg	Package
3DPM0237	DDR2	1.6 to 2.8 V	± 1A	50	80	SOP 24
3DPM0318	DDR2	1.6 to 2.8 V	± 1A	50	62.5	BGA 127
3DPM0385	DDR3	1.2 to 1.65 V	± 2A	75	62.5	BGA 259
3DPM0424	DDR4	1.1 to 1.3 V	± 2A	75	62.5	BGA 259



FLIGHT HERITAGE

With more than 175,000 modules in orbit as of 2021 and a failure-free flight heritage of more than 20 years, 3D PLUS is the largest Space Qualified multi-chip module manufacturer in Europe. 3D PLUS products are used by all the major space agencies and customers worldwide including commercial, scientific and governmental missions.

They bring key advantages to all space application fields and fly in numerous international missions:



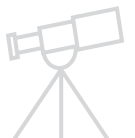
TRANSPORTATION

Orion, Ariane 5, HII-A/B, Space Launch System (SLS), COTS, Dream Chaser



CONSTELLATION

Iridium, Globalstar, OneWeb



REMOTE SENSING AND EARTH OBSERVATION

Goes-R, KhalifaSat, Cartosat, ICON, FormoSat-5, Sentinel, Pleiades, DubaiSAT-2, GEO-Kompsat 2A / 2B



NAVIGATION

Galileo, Glonass, Beidu



TELECOMMUNICATION

AlphaSat, Eutelsat W3C, Türksat 5A

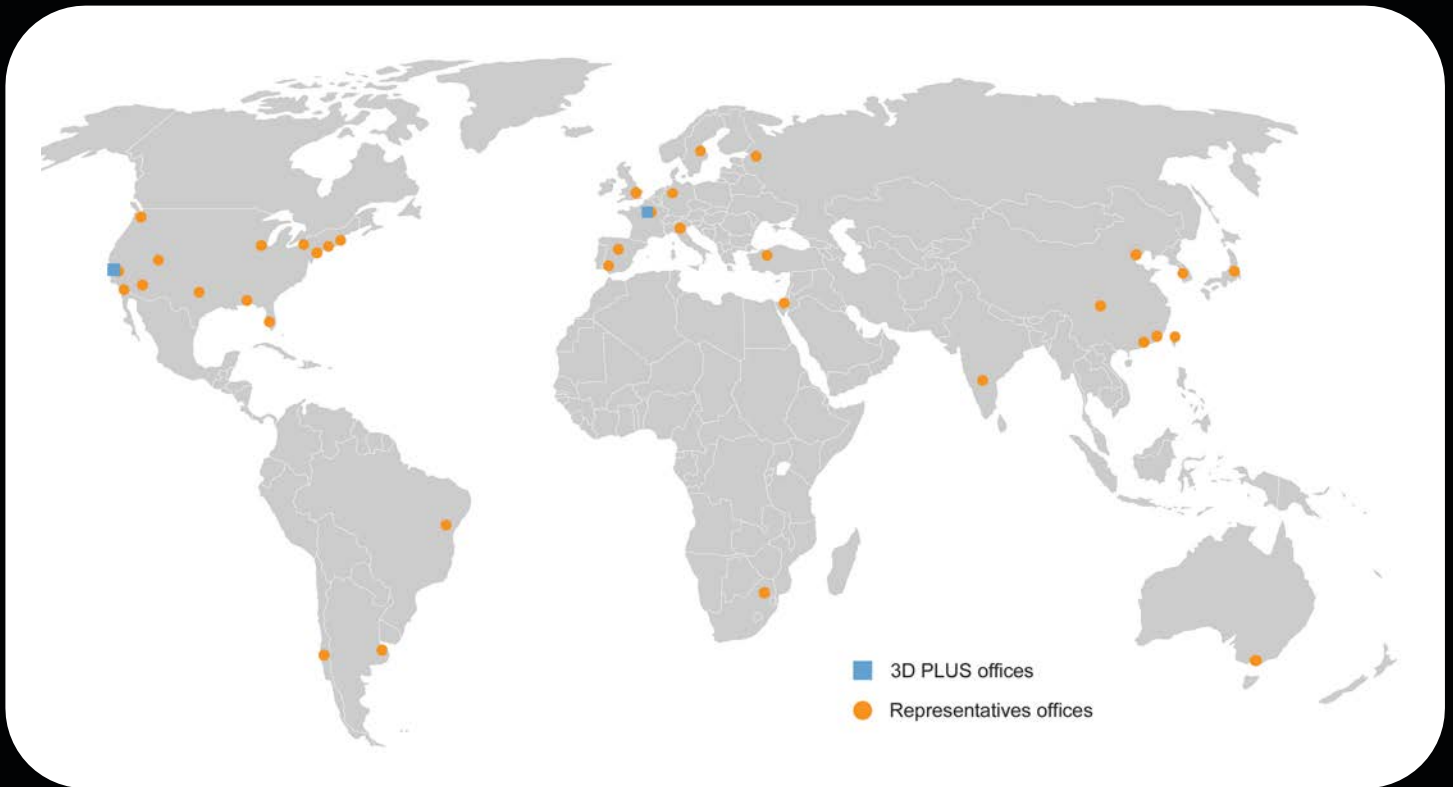


SCIENCE

Exomars 2016, Gaia, GOSAT, Kepler, Juno, OSIRIS REx, ASIM / ISS, Parker Solar Probe, New Horizons, Mars Science Laboratory, Insight, Rosetta, BepiColombo, Proba 2, Chang'e, Solar Orbiter, Mars 2020, Emirates Mars Mission, Tianwen-1



WE ARE **EVERYWHERE** YOU ARE



HEADQUARTERS (FRANCE)

408 rue Hélène Boucher
78530, Buc

Phone: +33 1 30 83 26 50
E-mail: sales@3d-plus.com



RELIABLE MINIATURIZATION TECHNOLOGIES FOR ELECTRONICS

www.3d-plus.com